

Analysis of Variance

ANOVA

ANALYSIS OF VARIANCE (ANOVA)

- ❑ Analysis of variance (Abbreviated as ANOVA)
- ❑ Variance is defined as the expectation of the squared deviation of a random variable from its mean i.e. S^2 or σ^2 .
- ❑ Analysis of Variance (ANOVA) is a method for testing the hypothesis that there is no difference between two or more population means.
- ❑ The ANOVA technique enables us to perform the simultaneous test and as such is considered to be an important tool of analysis.
- ❑ The significance of the difference of means of the two samples can be judged through either z-test or t-test.

- The technique of the analysis of variance was extremely useful in all types of researches.
- The variance of analysis studies the significance of the difference in means by analysing variance.
- The variance would differ only when the means are significantly different.
- H0: Variability within group = Variability b/t group,
H1: Variability within groups $\neq$ Variability b/t groups,

- The ANOVA technique was first developed by **Ronald A. Fisher**, a **British Scientist** in **1923**.
- For this technique **Fisher** is called the **Father of Modern Statistics**.
- It is the most powerful technique available in the field of statistical teaching.
- It is widely used in the experiments of behavioural and social sciences to test the significance of differences of means in different groups of a varied populations.
- Through this technique, it is possible to determine the significance of difference of different means in a single test rather than many.
- It minimizes the Type I error unlike in case of t- test.

Definitions of ANOVA

- “The analysis of variance is essentially a procedure for testing the difference between different groups of data for homogeneity”.

Yule and Kendal

- “Analysis of variance is the separation of the variance ascribe to one group of cause from the variance ascribe to other group.”

R.A. Fisher

Basic Principle of ANOVA

The basic principle of ANOVA is to test for differences among the means of the populations by examining the amount of variations within each of this samples,, relative to the amount of variation between samples.

Assumptions of Analysis of Variance

- Populations are normally distributed
- Population have same or equal variances
- Independent random samples are drawn

Classification

- One Way ANOVA
- Two Way ANOVA

One Way ANOVA

- The one-way analysis of variance is used to test the claim that three or more population means are equal
- This is an extension of the two independent samples t-test
- The one-way analysis of variance (ANOVA) is used to determine whether there are any significant differences between the means of three or more independent (unrelated) groups.

- The null hypothesis is that the means of groups are all equal.
- The alternative hypothesis is that at least one of the means is different.
- The ANOVA doesn't test that one mean is less than another, only whether they're all equal or at least one is different

- **dependent variable should be continuous.**
- **independent variable should consist of two or more categorical, independent groups.**
- We should have **independence of observations**, which means that there is no relationship between the observations in each group or between the groups themselves.

- There should be **no significant outliers**
- **Dependent variable** should be **approximately normally distributed for each category of the independent variable**
- There needs to be **homogeneity of variances**.

One-way classification

Let us suppose that N -observations y_{ij} ($i=1, 2, \dots, k$; $j=1, 2, \dots, n_i$) of a random variable y are grouped, on some basis, into k classes of sizes $n_1, n_2, \dots, n_k$ respectively, ($N = \sum_{i=1}^k n_i$) as exhibited in the table :-

one-way classified data

class	sample observations			Total	Mean
1	y_{11}	$y_{12} \dots y_{1n_1}$		T_1	$\bar{y}_1$
2	y_{21}	$y_{22} \dots y_{2n_2}$		T_2	$\bar{y}_2$
3	y_{31}	$y_{32} \dots y_{3n_3}$		T_3	$\bar{y}_3$
$\vdots$	$\vdots$			$\vdots$	$\vdots$
i	y_{i1}	$y_{i2} \dots y_{in_i}$		T_i	$\bar{y}_i$
$\vdots$	$\vdots$			$\vdots$	$\vdots$
k	y_{k1}	$y_{k2} \dots y_{kn_k}$		T_k	$\bar{y}_k$

The total variation in the observation y_{ij} can be split into the following two components :

- (i) The variation 'between the classes' $\Rightarrow$ Treatments
- (ii) The variation 'within the classes' $\Rightarrow$ the inherent variations of the random variables within the observations of a class.

* The first type of variations 'between the classes' is due to assignable causes which can be detected and controlled by human endeavours and the second type of variation is due to chance causes which are beyond the control of human hand.

ANOVA Table

For one-way classified data:-

Sources of Variation	Sum of Squares	degree of freedom	Mean Sum of Squares	Variance Ratio
Treatment (Ratio)	S_t^2	$k - 1$	$s_t^2 = \frac{S_t^2}{(k - 1)}$	$F = \frac{s_t^2}{s_E^2} = F_{k-1, N-k}$
Error	S_E^2	$N - k$	$s_E^2 = \frac{S_E^2}{(N - k)}$	"
Total	S_T^2	$N - 1$		

Q. A manufacturing company has purchased three new machines of different makes and wishes to determine whether one of them is faster than the others in producing a certain output. Five-hourly production figures are observed at random from each machine and results are given by Table. Use ANOVA technique and determine whether the machines are significantly different in their mean speed. $\alpha = 5\%$

Table

	Machine A_1	Machine A_2	Machine A_3
Observations	25	31	24
	30	39	30
	36	38	28
	38	42	25
	31	35	28

Solⁿ. The factor of variation is $\rightarrow$ Machines (A_1, A_2, A_3)

We setup the Hypothesis:-

Null Hypothesis: $H_0: \mu_1 = \mu_2 = \mu_3$: all the machines are equally effective.

H_1 : At least two of the means are not equal.

level of significance $\alpha = 0.05$

given $n_1 = n_2 = n_3 = 5$, $k = 3$, $N = n_1 + n_2 + n_3 = 15$

calculations For various Sum of Squares (S.S.)

Machine	sample observations (y_{ij})					Total	
A_1	25	30	36	38	31	$T_1 = 160$	$T_1^2 = 25,600$
A_2	31	39	38	42	35	$T_2 = 185$	$T_2^2 = 34,225$
A_3	24	30	28	25	28	$T_3 = 135$	$T_3^2 = 18,225$
Total $G = \sum \sum y_{ij} = 480$							$\sum T_i^2 = 78,050$

Sum of squares (S.S.)

$$\text{Raw S.S. (R.S.S.)} = \sum y_{ij}^2 = 25^2 + 30^2 + 36^2 + 38^2 + 31^2 + \dots + 25^2 + 28^2 \\ = 15,810$$

$$G = \sum \sum y_{ij} = 160 + 185 + 135 = 480$$

$$\text{correction factor (C.F.)} = \frac{G^2}{n} = \frac{(480)^2}{15} = \frac{230400}{15} = 15360$$

$$\text{Total S.S. (T.S.S.)} = S_T^2 = \text{R.S.S.} - \text{CF} \\ = 15,810 - 15,360 = 450$$

$$\text{Treatment S.S. (S}_t^2) = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - \text{CF} \\ = \frac{70,050}{5} - 15,360 = 250$$

$$S_t^2 = 250$$

$$\begin{aligned} \text{Error S.S. } (S_E^2) &= T.S.S. = S.S.T \\ &= 450 - 250 = 200 \end{aligned}$$

ANOVA TABLE

Source of variation (1)	Sum of squares (2)	d.f. (3)	Mean S.S. (4) = (2) ÷ (3)	Variance Ratio F
Treatment (Machines)	250	$(k-1)$ $3-1 = 2$	$\frac{S_t^2}{k-1} = \frac{250}{2} = 125$	$F = \frac{125}{16.67}$
Error	200	$N-k = 15-3$ $= 12$	$\frac{S_E^2}{N-k} = \frac{200}{12} = 16.67$	$= 7.49$
Total	450	$N-1 = 15-1 = 14$		

The F-value from Table

$$F_{\text{for } (k-1, N-k)} = F(2, 12) \text{ d.f. and } \alpha = 0.05 \\ = 3.89$$

$$F_{\text{cal}} > F_{\text{tab.}}$$

$\Rightarrow H_0$ is rejected $\Rightarrow$ Machines A_1, A_2, A_3 differ significantly.

Since H_0 of significantly rejected
equality of means rejected. $=$

ANOVA Table

Source	SS	DF	MS	F	P
Between SSG					
Within SSE					
Total SST					